

System Architecture

FitBuddy follows a modular FastAPI-based architecture.

Main Layers

Frontend Layer

HTML, CSS and Jinja2 templates are used for user interaction.

Backend Layer

FastAPI manages routes, form data and application logic.

AI Layer

Google Gemini models are used for workout generation, nutrition tips and feedback-based plan updates.

Database Layer

SQLite with SQLAlchemy is used to store user information and workout plans.

Architecture Flow

User



HTML / Jinja2



FastAPI Backend



Gemini AI



Workout / Nutrition Response



SQLite Database



Result Page

The project source specifies the same frontend, backend, AI and database architecture.